Lab 08

Goal

Create and resolve a real merge conflict across two files, where the correct fix is to combine both changes — not just pick one side.

Part 1: Diverge on purpose (local changes)

- List branches and switch into feature01

```
git branch
git switch feature01
```

- In Main.java, replace the addTask method so it also accepts a priority, while keeping the original one-argument version working by having it call the new one:

```
public static void addTask(String taskName) {
    addTask(taskName, "normal");
}

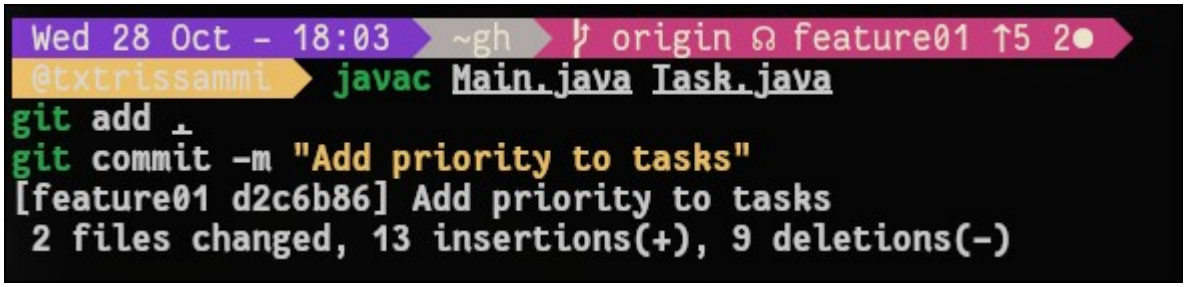
public static void addTask(String taskName, String priority) {
    tasks.add(new Task(taskName));
    System.out.println("Added task: " + taskName + " (priority: " + priority
+ ")");
}
```

- Also edit README.md and add a new bullet under your Features list:

```
- Assign a priority to each task
```

- Compile, save the files, add and commit the changes (do NOT push yet)

```
javac Main.java Task.java
git add .
git commit -m "Add priority to tasks"
```



Part 2: Meanwhile, on GitHub (a different change to the same lines)

- Go into the “tasktracker” repo in GitHub (main branch) and manually edit Main.java directly on GitHub: replace the SAME addTask method, but to support a due date instead:

```
public static void addTask(String taskName) {
    addTask(taskName, "none");
}

public static void addTask(String taskName, String dueDate) {
    tasks.add(new Task(taskName));
    System.out.println("Added task: " + taskName + " (due " + dueDate + ")");
}
```

- Also edit README.md directly on GitHub and add a bullet in the same spot in your Features list:

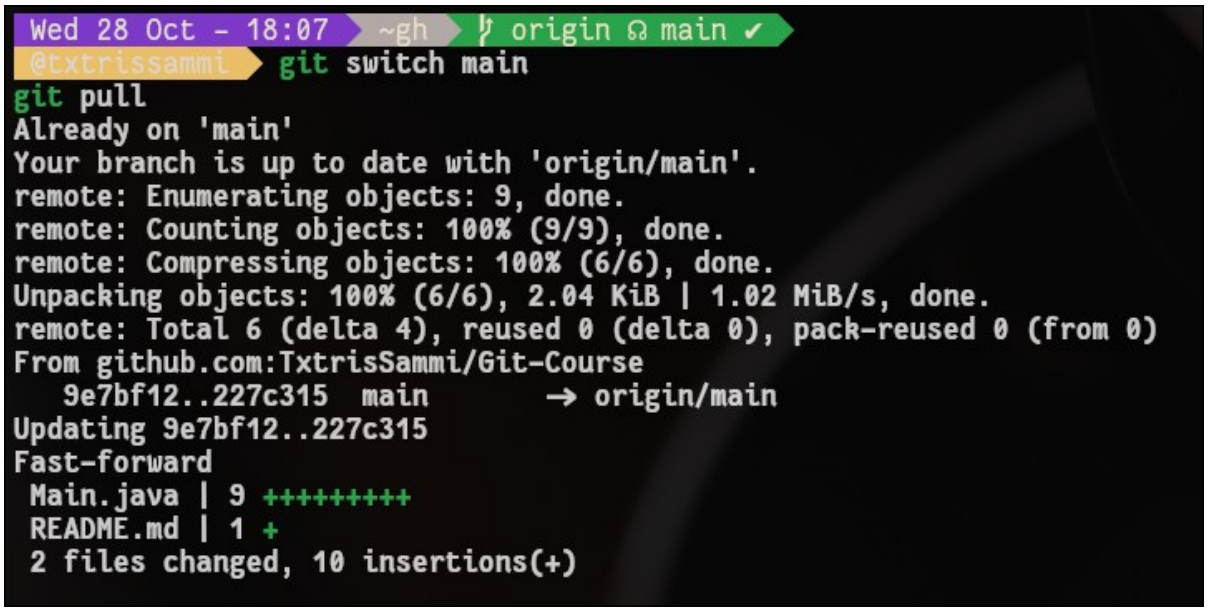
```
- Assign a due date to each task
```

- Commit both changes directly on GitHub's main branch

Part 3: Collide

- Back in git bash, switch to main and pull the changes

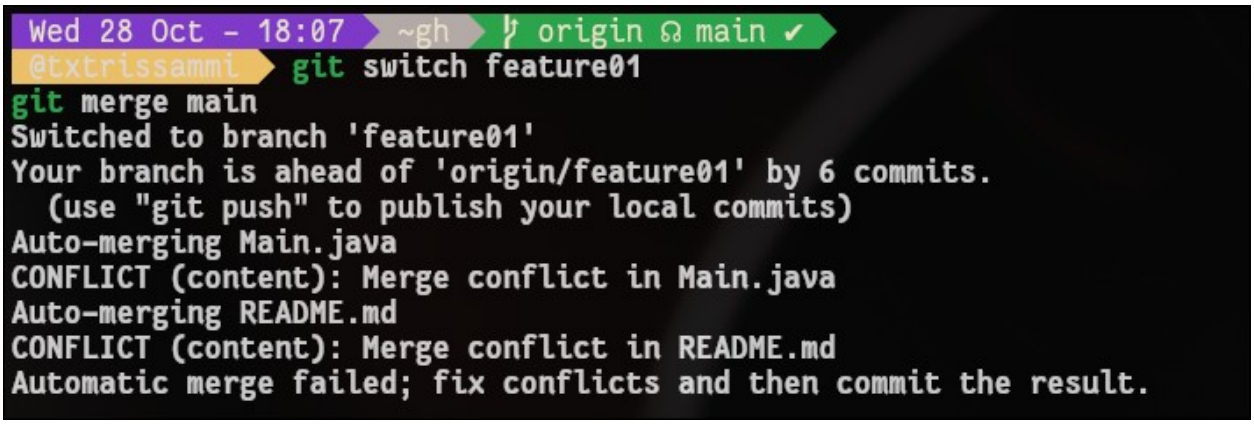
```
git switch main
git pull
```



- Switch back to feature01 and merge in main

```
git switch feature01
git merge main
```

NOTE: You should see TWO conflicting files this time: Main.java and README.md — both changed the same lines in two different ways.



Part 4: Resolve by combining both changes, not just picking one

- Open Main.java and rewrite addTask so it supports BOTH a due date and a priority, instead of choosing one side of the conflict. Keep the original one-argument version working by delegating to the new one:

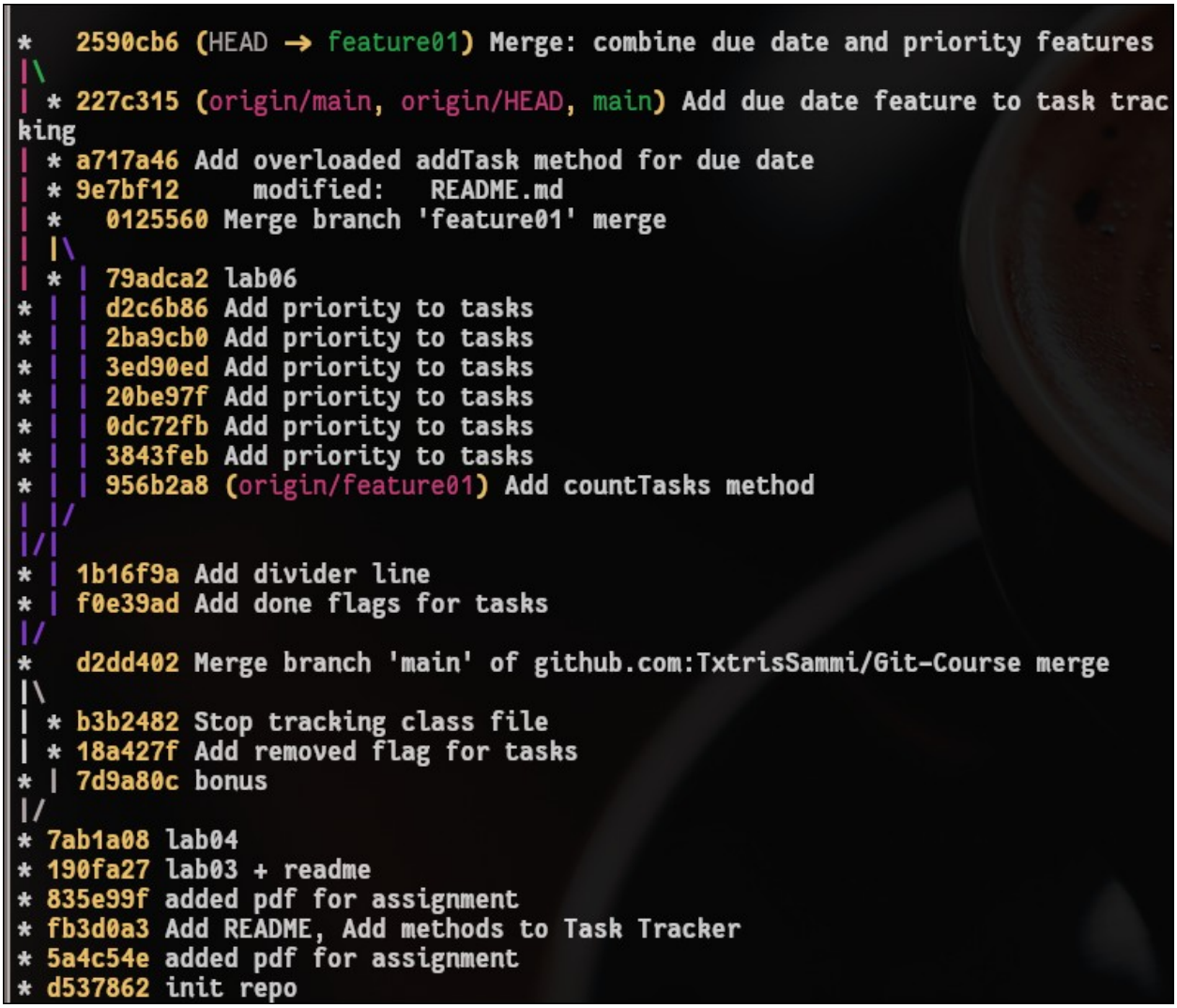
```
public static void addTask(String taskName) {
```

```
    addTask(taskName, "none", "normal");
}

public static void addTask(String taskName, String dueDate, String priority)
{
    tasks.add(new Task(taskName));
    System.out.println("Added task: " + taskName + " (due " + dueDate + ",
priority: " + priority + ")");
}
```

- Open README.md and keep BOTH new bullets (priority and due date) instead of deleting either one
- Remove the conflict markers, recompile to make sure everything still builds, then add and commit the resolved merge

```
javac Main.java Task.java
git add .
git commit -m "Merge: combine due date and priority features"
git log --graph --oneline --all
```



Submitting your solution

- Upload a PDF with screenshots of each step above, including the conflicting file contents before you resolved them and the combined result after
- Use the following Naming convention: Lab08_[Last Name].pdf

Example: Lab08_Maxwell.pdf